

# Engineering Projects in Community Service

## EPICS

### PROJECT SUMMARY FORM

#### SECTION 1 — Team & Academic Details

|                     |                                          |
|---------------------|------------------------------------------|
| Programme           | Computer Science & Engineering – AI & ML |
| Guide / Mentor Name | Mrs. M. Vani Pujitha                     |
| Guide Designation   | Assistant Professor                      |
| Guide Email ID      | pujitha@vrsiddhartha.ac.in               |
| Guide Contact No.   | 8074809058                               |

#### Team Members

| S.No | Student Name  | Roll Number | Email ID              | Contact No. |
|------|---------------|-------------|-----------------------|-------------|
| 1    | M. Kavya      | 24EU02030   | 24eu02030@vrsec.ac.in | 8096312030  |
| 2    | N. Bhagya sri | 24EU02038   | 24eu02038@vrsec.ac.in | 7382502324  |
| 3    | M. Harini     | 24EU02029   | 24eu02029@vrsec.ac.in | 9100210751  |

#### SECTION 2 — Project Domain & Client Details

|                         |                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------|
| Project Title           | AI-Assisted Early Grain Spoilage Prediction and Smart Storage Safety Monitoring System |
| Domain                  | Agriculture / Food Safety / Smart Monitoring                                           |
| Sub-Domain / Area       | Smart Grain Storage Monitoring                                                         |
| Client / Organisation   | Local Rice Mill and Grain Storage Units                                                |
| Client Contact Person   | M. Murali Mohan                                                                        |
| Client Designation      | Rice Mill Owner                                                                        |
| Client Email ID         | muralimandala74@gmail.com                                                              |
| Client Contact No.      | 9866802470                                                                             |
| Project Location        | Gudivada, Andhra Pradesh                                                               |
| Beneficiary Community   | Farmers, Rice Mill Owners, Grain Storage Operators                                     |
| Estimated Beneficiaries | 50–100 people initially                                                                |

### SECTION 3 — Problem Statement

|                                               |                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem Statement</b>                      | Grains stored in rice mills and storage warehouses often get spoiled due to excess moisture, temperature changes, fungal growth, insects, and poor storage conditions. Most storage systems are monitored manually, and spoilage is identified only after damage occurs, leading to food wastage and financial loss. |
| <b>Background / Context</b>                   | India stores huge quantities of grains such as rice and paddy in warehouses and rice mills. Improper storage conditions can damage grains and reduce quality. Farmers and storage operators face losses because spoilage is detected very late.                                                                      |
| <b>Root Cause Analysis</b>                    | <ul style="list-style-type: none"><li>• Excess humidity inside storage areas</li><li>• High temperature</li><li>• Poor ventilation</li><li>• Fungal growth and insects</li><li>• Lack of continuous monitoring</li><li>• Manual inspection methods</li></ul>                                                         |
| <b>Current Solutions (if any)</b>             | Existing systems mainly monitor temperature and humidity manually or using basic sensors. These systems only display environmental conditions and do not provide intelligent spoilage prediction or early warning alerts before grain damage occurs.                                                                 |
| <b>Justification for Engineering Approach</b> | Using IoT sensors and AI-based analysis can help continuously monitor storage conditions and predict spoilage risks at an early stage. This reduces grain loss, improves food safety, and helps storage operators take preventive action before major damage occurs.                                                 |

### SECTION 4 — Objectives

|                                                            |                                                                           |
|------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Objectives (List all specific goals of the project)</b> | <b>1. To continuously monitor grain storage conditions using sensors.</b> |
|                                                            | <b>2. To detect abnormal temperature, humidity, and gas conditions.</b>   |
|                                                            | <b>3. To predict spoilage risk using AI-based analysis.</b>               |
|                                                            | <b>4. To generate early alerts for storage operators.</b>                 |
|                                                            | <b>5. To reduce food wastage and financial loss.</b>                      |

| SECTION 5 — Expected Outcomes & Impact                 |                                                                                                                                                                                                     |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected Outcomes (Tangible / Measurable deliverables) | 1. Smart monitoring system for grain storage.                                                                                                                                                       |
|                                                        | 2. Early spoilage risk prediction.                                                                                                                                                                  |
|                                                        | 3. Real-time alerts and notifications.                                                                                                                                                              |
|                                                        | 4. Improved grain safety and reduced wastage.                                                                                                                                                       |
| Social Impact                                          |                                                                                                                                                                                                     |
| Short-Term Impact (Within project duration)            | <ul style="list-style-type: none"> <li>Real-time monitoring of storage conditions.</li> <li>Early alerts during abnormal conditions.</li> <li>Better awareness about safe grain storage.</li> </ul> |
| Long-Term Impact (Post deployment)                     | <ul style="list-style-type: none"> <li>Reduction in grain wastage.</li> <li>Improved food quality and storage management.</li> <li>Better support for farmers and storage operators.</li> </ul>     |
| Scalability Potential                                  | Yes. The system can be expanded to multiple rice mills, warehouses, and grain storage centers using IoT-based monitoring.                                                                           |
| Sustainability Plan                                    | The system can be maintained by storage operators using low-cost sensors and regular monitoring. Sensor modules can be replaced easily when needed.                                                 |

| SECTION 6 — Prototype Details |                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prototype Type                | <input type="checkbox"/> Hardware <input type="checkbox"/> Software <input type="checkbox"/> Hardware + Software                                                                                    |
| Prototype Description         | A smart monitoring system will be developed using sensors and ESP32 to monitor grain storage conditions. The system predicts spoilage risks and sends alerts when abnormal conditions are detected. |
| Key Technologies / Tools      | ESP32, IoT Sensors, Arduino IDE, Python, AI/ML                                                                                                                                                      |
| Programming Languages         | Python, Embedded C                                                                                                                                                                                  |
| Hardware Components           | <ul style="list-style-type: none"> <li>ESP32</li> <li>Temperature Sensor</li> <li>Humidity Sensor</li> <li>Gas Sensor</li> <li>Buzzer</li> <li>LCD Display</li> <li>WiFi Module</li> </ul>          |
| Software Modules              | <ul style="list-style-type: none"> <li>Sensor Data Monitoring</li> <li>Spoilage Prediction Module</li> <li>Alert Generation System</li> <li>Dashboard / Display Interface</li> </ul>                |
| Development Platform / IDE    | Arduino IDE, VS Code                                                                                                                                                                                |

|                          |                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Working Principle</b> | Sensors continuously monitor temperature, humidity, and gas conditions inside grain storage areas. The collected data is analyzed using AI-based logic to identify spoilage risk patterns. If abnormal conditions are detected, alerts are generated for storage operators. |
| <b>Prototype Stage</b>   | Concept / Design                                                                                                                                                                                                                                                            |
| <b>Demo Mode</b>         | Physical Demo                                                                                                                                                                                                                                                               |

## SECTION 7 — Project Plan & Resources

|                                    |                                                                                                                                                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Start Date</b>          |                                                                                                                                                                                                                                                                                      |
| <b>Expected Completion Date</b>    |                                                                                                                                                                                                                                                                                      |
| <b>Estimated Budget (₹)</b>        | ₹3000 – ₹5000                                                                                                                                                                                                                                                                        |
| <b>Funding Source</b>              | Self-funded                                                                                                                                                                                                                                                                          |
| <b>Milestones / Timeline</b>       | <ul style="list-style-type: none"> <li>• Problem analysis and requirement collection.</li> <li>• Sensor selection and hardware setup.</li> <li>• Software and AI module development.</li> <li>• Integration and testing.</li> <li>• Final prototype demonstration.</li> </ul>        |
| <b>Risks &amp; Mitigation Plan</b> | <ul style="list-style-type: none"> <li>• Sensor failure → Replace sensors and recalibrate</li> <li>• Incorrect predictions → Improve dataset and testing</li> <li>• Connectivity issues → Use local monitoring backup</li> <li>• Power interruptions → Use battery backup</li> </ul> |

## SECTION 8 — Declarations & Signatures

**Declaration:** We, the undersigned team members, hereby declare that the information provided in this application is true and accurate to the best of our knowledge. We understand that the project must genuinely serve the community and adhere to the EPICS programme guidelines.

| <b>Team Member Name</b> | <b>Roll No.</b> | <b>Signature</b> | <b>Date</b> |
|-------------------------|-----------------|------------------|-------------|
| M. Kavya                | 24EU02030       |                  |             |
| N. Bhagya sri           | 24EU02038       |                  |             |
| M. Harini               | 24EU02029       |                  |             |